October 21, 2026

Journal of Machine Learning Research

Dear Editors:

We are writing to submit our manuscript "torchtsetlin: Fast, Interpretable Tsetlin Machines for the PyTorch Ecosystem" to the *Journal of Machine Learning Research*, for the **Machine Learning Open Source Software (MLOSS)** track.

Our manuscript describes `torchtsetlin`, an open-source library that implements the Tsetlin machine family (vanilla, weighted, coalesced, regression, convolutional over sequences, images and volumes, and dense segmentation) as ordinary PyTorch modules. Tsetlin machines learn human-readable propositional rules without gradients, but their software has so far consisted of C and CUDA kernels that sit outside the deep-learning ecosystem. `torchtsetlin` stores the automata as tensors, so models move between devices, serialise and consume data loaders like any other PyTorch model, and it introduces a batched feedback algorithm that turns per-example automaton updates into matrix products and binomial draws. This makes Tsetlin machines practical on GPUs (**up to $X\times$** faster than the same code on a multi-core CPU) and easy to extend in Python. The library is documented, tested on CPU and GPU, continuously integrated, and released on PyPI.

- **Open source licence:** MIT.
- **Project web address:** `https://github.com/vlbthambawita/torchtsetlin` (documentation: `https://vajira.info/torchtsetlin/`; PyPI: `torchtsetlin`).
- **Software version to be reviewed:** **0.3.x, git tag / commit hash**.
- **Evidence of an active user community:** **PyPI downloads, GitHub stars/forks, issues and pull requests from external users, dependent projects, citing works**.

This manuscript overlaps with our technical report **arXiv identifier**, which is a non-peer-reviewed preprint; **state how this submission differs, or that it is a condensed version**.

We suggest the following action editors and referees for our submission.

Action Editors:

- **First Lastname, Institution (email)**
- **First Lastname, Institution (email)**
- **First Lastname, Institution (email)**

Reviewers (with a brief note on relevance):

- **First Lastname, Institution (email): relevance**
- **First Lastname, Institution (email): relevance**
- **First Lastname, Institution (email): relevance**

Our submission has the following keywords: Tsetlin machine, interpretable machine learning, propositional logic, PyTorch, GPU.

As the corresponding author, I confirm that none of the co-authors listed below have a conflict of interest with the action editors and referees I suggest above. Further, I confirm that all co-authors below consent to my submission of this manuscript to the *Journal of Machine Learning Research*.

Sincerely,

Vajira Thambawita (Simula Metropolitan Center for Digital Engineering)
Co-Author (Institution)